11th of December the chart was filled with soundings ranging between nine feet and twenty-three fathoms.

In order to join with the inshore soundings last mentioned, a fifth sheet was projected for extending the hydrography southward to Point Sal, where operations were closed for the season on the 11th of December. The aggregate statistics are:

Miles run in sounding	642
Angles measured	1,838
Number of soundings	7,676

Lieuts. E. K. Moore and L. C. Heilner, U. S. N., and Masters W. P. Elliot and R. H. Galt, U. S. N., were attached to the hydrographic party in the steamer McArthur.

Topography north of Point Arguello, Cal.—After completing the tertiary triangulation of the coast of California between Point Arguello and Point Sal, as stated in my report of last year, Assistant W. E. Greenwell closed that season by a detailed topographical survey of the middle stretch of coast included in the triangulation. The sheet returned to the office represents all the topographical features adjacent to Point Purissima. Of three other projections filled in the course of the present fiscal year one shows the shore line and surface details along the coast from Arroyo Hondo to Lompoc Landing, which is a short distance south of Point Purissima. The sheet of last year including that point delineates nine miles of the coast northward to Shuman's Cañon. There the plane-table survey was resumed, and during September, October, and November, 1879, was prosecuted northward around Point Sal to the entrance of Santa Maria River. Later in the year a third sheet was filled with details extending the topographical survey to Arroyo Grande, where a junction was made with the plane-table work formerly done on the shores of the San Luis Obispo Bay. The three sheets turned in by Assistant Greenwell represent the ground features to a distance of about two miles from the coast line. A synopsis of the general statistics is appended:

Shore line surveyed, miles	25½
Roads, miles	55
Area of topography, square miles	55

Primary triangulation.—For completing observations at the eastern end of the great quadrilateral which so appropriately bears his name, Prof. George Davidson, Assistant in the Coast and Geodetic Survey, organized his party and took the field near the end of May, 1879, after arranging to occupy first Mount Lola (ninety-two hundred and eighty feet high) and next Round Top, the summit of which is ten thousand six hundred and forty-five feet above the sea level. The route to the last-named station was greatly impeded by snow until late in June.

"Mount Lola is about twelve miles north-northeast of Summit Station, on the Central Pacific Railroad, and is not specially distinguishable for elevation. It is on the main crest line of the Sierra Nevada, and may be reached from the eastward by several ways; but going by way of Lake Independence the party had only fourteen miles of teaming in a rise of twelve hundred feet. The summit was reached by packing and sledding five miles and a half over snow, with a rise of about twenty-eight hundred feet."

Much difficulty was encountered in consequence of drifts, but the party was thoroughly efficient and met all impediments successfully. While Assistant J. J. Gilbert and Subassistant E. F. Dickins advanced towards the station with the instruments and needful equipage, Assistant B. A. Colonna was detailed by Professor Davidson to post heliotropers at Mount Helena, Snow Mount, Mount Linn, and Marysville Butte. At the same time, Subassistant J. F. Pratt was sent to station heliotropers at Mount Diablo, Pine Hill, and Lessen's Butte. Near the last-named station the snow was very deep, but Mr. Pratt succeeded in reaching it and posting the hand engaged as heliotroper.

Much hardship was incurred in reaching Mount Lola, on account of snow and cold air. Officers, men, and animals sank in the snow to their knees, and all were severely tried in the ascent, which of necessity was done afoot in a rare atmosphere. Assistant Davidson went to the summit on the 9th of June, and then encountered a flurry of snow and found the temperature below the freezing point. As soon as possible, a space was cleared by the removal of detached rocks, and a pier was constructed by using cement. This was properly capped to receive the theodolite, and

was marked for identifying the station point. In lieu of a wooden observatory, Assistant Davidson trusted to his canvas tent, and protected it by a heavy dry wall of stone which broke the force of many fierce winds in the course of the season. When the pier was ready, Subassistant Dickins was sent to post heliotropers at Carson Cone and Pah-Rah. Subassistant Pratt was detailed to proceed to Round Top and report upon the best means of reaching that summit and in regard to the subsistence of the party while there. Professor Davidson selected an azimuth mark somewhat more than five miles distant from Mount Lola, and in so doing guarded as far as possible against the effect of horizontal refraction.

After mounting the theodolite, observations for horizontal directions were commenced without delay, and while these were conducted by Assistant Davidson two piers were built by Assistants Colonna and Gilbert for the transit and zenith telescope. The foundations were of rock laid in cement; the superstructure of brick capped with a flat stone. The vertical circle was mounted on a very heavy block of pine, and another block was arranged to receive the magnetometer. For horizontal directions and azimuth, the twenty-inch theodolite No. 115 was mounted on the cast-iron repeating stand devised by Assistant Davidson in 1870. Observations were recorded in twenty-three positions of the limb of the instrument, and the readings were made by three micrometer microscopes with circle, direct and reversed, upon two hundred and seventy-six different graduations (of five minutes each) upon the limb.

Assistant Davidson, having stationed Assistant Colonna at Mount Shasta, succeeded in measuring two angles in the great triangle Lola-Shasta-Helena, having sides of one hundred and thirty-six, one hundred and sixty-nine, and one hundred and ninety-two miles. Under a subsequent heading, mention will be made of the operations upon Shasta. On the 29th of July, Mr. Davidson saw the heliotrope shown from that station, and observed upon it in five positions of the theodolite at Lola. Further tests were made there for the determination of azimuth by the use of a collimator, but the results served only to establish the conclusion reached previously, and reported by Professor Davidson, "that the collimator, to be effective, must be upon the same rock, or upon a pier connected in solidity with the pier of the theodolite." These conditions were impracticable at Mount Lola.

For azimuth, Assistant Davidson observed on the two stars, α Ursæ Minoris at eastern elongation, and upon B. A. C. 4165 at western elongation, in the twenty-three positions of the instrument. The observations were referred to the azimuth mark and also to the collimator. While this work was in progress Subassistant Dickins kept the record.

The latitude observations, and partly those for time, were made by Assistant Colonna, with zenith telescope No. 3. Twenty-five pairs of stars were observed through seven nights, and the usual record was made for determining the value of the micrometer screw, and for measuring the irregularity of each turn.

After the departure of Mr. Colonna for Mount Shasta, time observations with the transit were continued by Mr. Gilbert. In the measurement of vertical angles full series were recorded at Mount Lola upon all the signals in view, one of them being one hundred and sixty-nine miles distant.

Magnetic observations were made at the summit, but, as remarked in the field report, they are to be considered only a tentative for absolute measures. Later in the season, Professor Davidson set up the theodolite magnetometer at Lake Independence, which is twenty-one hundred feet below the summit of Mount Lola, and there Mr. Dickins observed for the magnetic declination during one day, while Assistant Davidson observed for time and azimuth. Dense smoke in the Sacramento Valley retarded operations at Mount Lola, but, pending the delay, a topographical reconnaissance of the mountain was made by Subassistant Dickins. For the resulting map, Assistant Davidson selected and measured a base, and made sketches of the characteristic features. Subsequently photographic views were secured of important points to the southward and eastward, in which directions smoke did not interfere as it did in other directions. Meteorological observations were recorded at Lola in July, and in August and September at Round Top. As before stated, the heliotrope on Mount Shasta was seen on the 29th of July, when all the baggage on Lola was already packed for the transfer to Round Top. There Mr. Pratt provided huts by the 9th of August, and had located a wagon road to the lower camp, as also a trail to the observing point

near the summit of the mountain. Lumber, camp equipage, instruments, wood, water, and provisions were of necessity carried up to the station. There the pier for the large theodolite was built by Subassistant Pratt himself, of loose rocks at the summit, which he laid in cement. As before, the instrument was mounted on Mr. Davidson's cast-iron "position circle." The little wooden observatory, eight feet square, was protected from blowing away by a heavy dry wall of rocks, and holes with slides were cut in the lines leading to the different stations. Assistant Colonna built the piers for the zenith telescope and transit. Wherever needful, dry walls were put up for securing what had been taken to the summit, the elevation being ten thousand six hundred and forty-five feet, and the peak of the mountain swept by fierce winds.

As at Lola, horizontal angles were measured at Round Top under all circumstances of weather. Referring to the condition of the atmosphere late in September, Assistant Davidson says:

"I have stood hours at the instrument to get twenty minutes' work, and sometimes to get nothing. The cold was severe, and in the last snow, which was fifteen feet deep on the trail, we pulled ourselves up to the station by means of a life-line through snow breast high when the air temperature was near zero.

"In order to use at Round Top the same stars that had been observed on at Lola, the azimuth mark was set up by Subassistant Pratt at a point about six miles north-northwest of the station and seven hundred feet less in elevation."

Professor Davidson made the observations, and at the close assisted Messrs. Gilbert and Dickins in practice for observing azimuth.

For latitude, Assistant Colonna observed twenty-five pairs of stars on seven nights, and, as before, determined the value of the micrometer for an entire turn of the screw as well as for irregularity at each turn. He also made the transit observations, using the Davidson meridian instrument, the records being made by Mr. Dickins, who, after their completion, executed a topographical survey of the vicinity of the station on Round Top. Many difficulties were overcome in securing photographic views.

From Mount Lola, the lengths of the lines observed on are fifty-four, fifty-seven, fifty-eight, sixty-one, eighty-one, ninety-six, one hundred and thirty-one, one hundred and thirty-six, one hundred and thirty-eight, one hundred and forty-one, one hundred and sixty-nine, and one hundred and ninety-two miles, the last being to Mount Shasta. From Round Top the lines of triangulation range at thirty-eight, fifty-four, fifty-eight, sixty-seven, sixty-eight, sixty-eight, forty-five, one hundred and seven, one hundred and twenty, one hundred and twenty, one hundred and forty-six, and one hundred and sixty miles. At the two stations thirty-one hundred and forty-eight observations were recorded for determining the azimuth.

Latitude was determined at Mount Lola and Round Top by an aggregate of sixteen hundred and forty-four observations.

The meteorological records include upwards of two thousand entries.

At Lola, fourteen days and three nights were given to determine the coefficient of refraction, and sixteen days with two nights for the same purpose at Round Top. An aggregate of twenty-two hundred and ninety-three observations were thus recorded; and eighteen hundred and forty-six observations for the measurement of vertical angles. At Lola and at Independence Lake, Professor Davidson made about four hundred observations for determining the magnetic elements. The ordinary statistics for the measurement of horizontal angles at the two stations are:

Horizontal measures in twenty-three positions	3,310
Pointings of ocular micrometer	12,770
Primary signals observed on	23
Secondary signals observed on	40

At Jackson Butte, about forty-six miles southwest of Round Top, Assistant Davidson recorded about eleven hundred observations for determining the coefficient of refraction, and measured horizontal angles on two primary stations. Incidentally, Professor Davidson examined the region about Truckee, on the Central Pacific Railroad, with reference to the practicability of determining, by means of the spirit level, the difference in height between that point and the station on Mount Lola.

The report of Assistant Davidson gives the results of his observations on heliotrope spectra in remarks of special interest:

selected for conducting the primary triangulation of the coast of California from former limits as far northward as Point Gorda. At the opening of the present fiscal year the work had been advanced about one hundred and fifty miles, as reckoned by the coast line, above the entrance to San Francisco Bay. For its continuation, Assistant A. F. Rodgers, in the course of the season, occupied first the Great Caspar station near the coast, and subsequently Two Rock Peak. From these points nine signals were observed on, and from the last named vertical angles were measured for determining five elevations. Eight volumes, containing records of the horizontal and vertical angles observed at the two primary stations, have been received at the office. At the close of the fiscal year Assistant Rodgers was preparing to occupy Mount Sanhedrin for angular measurements to the northward. The statistics of the work done are:

Primary stations occupied	2
Angles measured	7
Number of observations	6,876

Subassistant D. B. Wainwright was attached to this field party, and also assisted in the computations.

Mount Shasta geodetic station.—Under a preceding head it was stated that a signal on Mount Shasta had been observed on from two stations of the great quadrilateral that crosses the State of California along the thirty-ninth parallel of latitude. From the two stations the distances to Shasta were approximately one hundred and sixty-nine and one hundred and ninety-two miles. The line joining the two stations was one hundred and thirty-six miles in length.

The desirable result just mentioned was effected by the steady endurance of Assistant B. A. Colonna, who passed nine days and nights on the summit of Shasta, the elevation of which is fourteen thousand four hundred feet above the sea level. During the ascent, and while on the summit, many collateral observations of interest were recorded and returned in the field report. At an elevation of about ten thousand feet, snow, white at the surface, was found to be red at a depth of two inches. The coloring matter extended to a depth of three inches, and a handful of the fungi, after evaporating the snow, had a decidedly fruity taste. From this elevation Mr. Colonna, accompanied by two men, started for the top of the mountain on the 25th of July, 1879. The instruments and requisite provisions were taken up in parcels by Indians hired for the occasion. Loose bowlders that would start down with a mere touch made the steep ascent somewhat dangerous, but accidents from such causes were carefully avoided. Under a steep wall of pumice, called "Red Bluff," as seen from the valley, the party, at the height of thirteen thousand feet, passed over a snow drift, part of which adjacent to the rock had melted, leaving a narrow chasm of which no bottom could be seen. There the rarefied air affected respiration, but by frequent halts this part of the journey was passed in safety, and at noon the party reached the hot springs, two hundred and fifteen feet below the summit of Shasta. All the carriers returned immediately to the valley. The two men retained by Assistant Colonna aided him in pitching a tent, but before night one of the men was taken ill and next morning went down the mountain and did not return. The record shows that after dark the thermometer stood at 30°.

As described by Mr. Colonna, the summit of Shasta consists of two peaks, about two hundred yards apart, one of them towards the northeast being about fifty feet higher than the other. The hot springs are in the valley between the two peaks, dotted over an area of nearly twenty yards square, and they constantly emit steam impregnated with sulphureted hydrogen and other gases. Assistant Colonna found the temperature of the water 184°, which, considering the elevation, is perhaps equal to the normal boiling point at sea level. The adjacent surface is a crust made up of loose earth and small stones cemented with sulphur, alum, and other substances. At one place the crust when broken through disclosed a cavern from which steam came out in great volumes; and from another break the steam was emitted as a small jet with a hissing sound like that caused by a locomotive. It was noticed that, although the temperature did not change, all the springs were more active on some days than on others.

Assistant Davidson had arranged for showing heliotropes, from the two stations Helena and Lola, of the great quadrilateral, but intervening smoke hid them from view at Shasta until the 29th of July, when, at $5^h\ 45^m$ a. m., Mr. Colonna "saw the Lola heliotrope for a second or two dark red,

but it went before a pointing could be recorded. Helena was not seen until the 1st of August, and then four sets of six repetitions were measured. The same measures were repeated next day, but on the 3d Helena was not visible." The line from Helena to Shasta (one hundred and ninety-two miles) is the longest yet included in geodetic work. As before stated, Assistant Davidson observed on the Shasta heliotrope both from Mount Lola and Helena.

The experience of many tourists who have ascended Mount Shasta proves that few men could endure even the temporary sojourn of nine days and nights at such elevation. Assistant Colonna retained his appetite and was amply supplied with provisions, yet his weight was lessened fifteen pounds by the stay on the mountain. He experienced no difficulty in breathing, but noticed that by a little exertion the pulse rapidly increased, going as high as one hundred and twenty, though in repose at the summit it ranged from ninety to one hundred.

During his stay the highest temperature observed was 67°, when at the same time the thermometer in the valley marked 100° in the shade. At four o'clock in the afternoon ice would form in the sun, and at sundown the thermometer was generally at 25°. The coldest temperature recorded at the summit was 18°. With strong winds the temperature increased as though warm air had been thereby forced up the side of the mountain.

After visiting the Whitney glacier and making many collateral observations at other points on Shasta, Assistant Colonna left the station and reached San Francisco on the 9th of August.

Tidal observations.—The series of tidal observations commenced in June, 1877, near the entrance to the harbor of Honolulu, under the direction of Mr. W. D. Alexander, superintendent of the survey of the Sandwich Islands, has been continued. Records from the beginning and down to the 18th of February, 1880, have been received and filed at the office in Washington. As heretofore mentioned, the gauge in use at Honolulu was lent from this office on condition that the registers should be furnished for comparison between the tides there and the tides of the Pacific coast of the United States.

SECTION XI.

OREGON AND WASHINGTON TERRITORY.—(SKETCHES NOS. 24 and 25.)

Topography of Columbia River.—At the opening of the fiscal year, Assistant Cleveland Rockwell was again assigned to continue the survey of the banks of the Columbia. The sheet projected includes the course of the river between Kalama and Columbia City, and as returned to the office shows more than the usual proportion of intricate details within a sketch of rather more than eight miles. Both banks of the river were carefully mapped in contour, and all the islands are represented, as well as the streams that make into the Columbia.

Field work by this party was continued until the 8th of December, 1879, when the sloop Kincheloe, which had served for transportation and quarters, was moored for the winter at the mouth of Kalama River in a position secure from floating ice. The party was then discharged and Mr. Rockwell engaged in office work. A synopsis appended to the field report shows in statistics:

Shore line surveyed, miles	16
Sloughs and ponds, miles	44
Roads, miles	38
Area of topography, square miles	24

By the work here noticed the detailed survey of the Columbia has been extended to a point about seventy-five miles above the mouth of the river.

Triangulation and topography, Puget Sound, W. T.—At the opening of the fiscal year, Subassistant Eugene Ellicott entered upon the triangulation of Hammersley's Inlet, and completed that work early in August, 1879. In August, the shore lines were traced of the lower part of Hartstein's Island, the shores of Squaxin Island, part of the shores of Totten Inlet, and the east side of the north end of Case's Inlet. The shores of Hammersley's Inlet and Skookum Inlet were surveyed in September, and Mr. Ellicott furnished tracings of the sheets to Lieutenant Wyckoff, whose party was then engaged in sounding the upper waters of Puget Sound.

The topographical work was much retarded by high tides. It frequently happened that